

MH1301 Discrete Mathematics

Handout 12: Review

Final Exam

- Coverage: everything from the lectures, except the slides that are marked “optional”.
- Weight
 - ▶ Counting: 35%
 - ▶ Recurrence: 30%
 - ▶ Graph Theory: 35%
- May 9th 9am-11am; Exam Hall A.
- Format: Restricted open book: One double-sided A4-size reference sheet with texts handwritten or typed on the A4 paper (no sticky notes/post-it notes on the reference sheet); calculators allowed; dictionary for foreigner students allowed;
- Lecturer will be around as invigilator.

Level of Difficulty

- Easy questions approx. 45 marks
 - Slightly more challenging questions 55 marks.
 - Targeted average marks: 40 (from easy questions) + 25 (from challenging questions) = 65 marks.
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- CA weighting: 40%. Average: 35%
- Final exam weighting: 60%. Target average: 39%.
- Target average for this course:

$$35\%(\text{CA}) + 65\% \times 60\%(\text{Final}) = 74\%.$$

Counting (Questions 1 & 2)

- Count the union of sets

- ▶ if sets are disjoint, **Sum Rule**

$$|A_1 \cup A_2 \cup \dots \cup A_n| = |A_1| + |A_2| + \dots + |A_n|$$

- ▶ otherwise, **Inclusion-Exclusion principle**

- * $|A_1 \cup A_2| = |A_1| + |A_2| - |A_1 \cap A_2|$

- * $|A_1 \cup A_2 \cup A_3| = |A_1| + |A_2| + |A_3| - |A_1 \cap A_2| - |A_1 \cap A_3| - |A_2 \cap A_3| + |A_1 \cap A_2 \cap A_3|$

- Product rule** for count sequences

- Division rule**

- Bijective function mapping rule:** if function $f : X \mapsto Y$ is a bijection, then $|X| = |Y|$.

Counting (Questions 1 & 2)

- Pigeonhole Principle
- Binomial theorem

$$(x + y)^n = \sum_{i=0}^n C(n, i)x^{n-i}y^i.$$

- **Combinatorial proof:** a proof that uses counting arguments to prove that both sides of the identity count the same objects but in different ways.

Counting (Questions 1 & 2)

- **Permutations** $P(n, r)$: number of **ordered** selections

$$P(n, r) = \frac{n!}{(n-r)!}$$

- **Combinations** $C(n, r)$: number of **unordered** selections

$$C(n, r) = \frac{n!}{r!(n-r)!} = \binom{n}{r}$$

Counting (Questions 1 & 2)

r -permutation

r -combination

w/o repetition

$$P(n, r) = \frac{n!}{(n-r)!}$$

$$C(n, r) = \frac{n!}{(n-r)!r!}$$

w/ repetition

$$n^r$$

$$C(n+r-1, r)$$

w/ indistinguishable
objects

$$\frac{n!}{n_1!n_2!\cdots n_k!}$$

around a cycle

$$\frac{P(n, r)}{r} = \frac{n!}{r(n-r)!}$$

Recurrence Relations (Questions 3 & 4)

Recurrence Relation

- For a sequence $\{a_n\}$, a recurrence relation is a formula that relates each term a_n to certain of its predecessors $a_{n-1}, a_{n-2}, \dots, a_{n-k}$, where $k \leq n$ is an integer.
 - ▶ recurrence formula
 - ▶ initial values
- Find the connection between problem of size n with those with size $n-1, n-2, \dots$

Recurrence Relations (Questions 3 & 4)

Solving Recurrence Relation

Linear homogenous recurrences

- write down characteristic equation
- find characteristic roots
- use initial conditions to determine the coefficients

Linear nonhomogenous recurrences

- solve its associated homogenous recurrence
- find a particular solution
 - ▶ guess the format of the particular solution
 - ▶ use recurrence relation to determine the coefficients

Graph Theory (Questions 5)

- **Graph:** definitions and terminologies
 - ▶ directed/undirected graph, vertex, edge, simple graph
 - ▶ neighbours, endpoints, degree
 - ▶ subgraphs
 - ▶ path, path length, simple path, cycle (circuit)
 - ▶ connected graph, connected component
 - ▶ edge-weighted graph
 - ▶ induced graph
 - ▶ $G - \{v\}$ or $G - v$
- **Handshaking Lemma:** relation between vertex degrees and number of edges in the graph
 - ▶ undirected and directed versions
- **Graph Isomorphism:** two graphs that “look the same” but have different labels for vertices

Graph Theory (Questions 5)

Special Graphs

- Complete graph K_n
- Cycle C_n
- Complete bipartite graph $K_{m,n}$

Graph Theory (Questions 5)

Euler circuit/path: a circuit/path that visits every **edge** in the graph exactly once.

- A connected graph has an **Euler circuit** if and only if all vertices have even degrees
- A connected graph has an **Euler path** if and only if 0 or 2 vertices have odd degrees

Hamiltonian circuit/path: a circuit/path that visits every **vertex** in the graph exactly once.

- no “easy” way to decide their existence in a graph

Graph Theory (Questions 5)

Graph Coloring: color each vertex in one of k different colors such that no edge connects two same-color vertices

- **chromatic number:** minimum number of colors needed to color a graph

Bipartite Graph: graphs that can be colored in two colors or less

- a graph is bipartite if and only if it does not contain any odd cycle

Graph Theory (Questions 5)

Planar Graph: a graph that **can be** drawn in a 2D plane without any edges crossing

- non-planar graphs: K_5 and $K_{3,3}$

Euler's Formula: $\# \text{region} = \# \text{edges} - \# \text{vertices} + 2$

- corollary 1: $m \leq 3n - 6$ for any planar graph
- corollary 2: $m \leq 2n - 4$ for any planar graph that doesn't contain C_3

Four-Color Theorem: every simple planar graph is 4-colorable.

Graph Theory (Questions 5)

Tree: definitions and terminologies

- tree, forest, leaf
- rooted tree, level, height, parent, child, ancestor, descendant

Binary Tree: every vertex has no more than 2 children

- full binary tree, balanced binary tree
- at most 2^h leaves when the tree height is h

Graph Theory (Question 6)

Spanning Tree: tree subgraph that connects all vertices

Minimum Spanning Tree: spanning tree with smallest total sum of edge weights

Two algorithms for finding MST:

- **Kruskal:** Process edges in the increasing order of their costs (break ties arbitrarily), and add edges to T as long as they don't form a cycle.
- **Prim:** T maintained by the algorithm will be a tree, starting from a single vertex. In each iteration, pick edges with least attachment cost to T .

Graph Theory (Question 6)

Shortest Path: find a path of shortest length from one vertex to another

- definition: path length in weighted graphs, distance

Dijkstra Algorithm: find distance from s to all other vertices

- maintain S : set of vertices closest to s
- maintain $d(u)$: current known shortest distance from s to u
- initially $S = \emptyset$, $d(s) = 0$, $d(v) = \infty$ for $v \neq s$
- at each step
 - ▶ choose $u \in V - S$ with smallest $d(u)$ to add to S
 - ▶ update every other vertex v adjacent to u
 $d(v) = \min\{d(v), d(u) + \ell(u, v)\}$

Graph Theory (Question 6)

let $a_{n(i)}$ be the string that does not contain ternary strings of length n and ends with i

$$a_n = a_{n-1}(0) + a_{n-1}(1) + a_{n-1}(2)$$

